

CLINOMA PLATFORM

Second Paper Camp — Day 2

Pediatrics Matching Quiz (Student Copy)

Topics: Nephrology • Neonatology

Duration: 30 Minutes • Total Questions: 30 (6 Matching Sets)

Prepared by: Clinoma Platform Team

Interactive Matching Sets Examination

Group 1: Clinical Presentations & Definitions (Nephrology)

A. Cystitis (UTI)	B. Stage 2 "Injury" (pRIFLE Criteria - AKI)
C. Nephrotic Syndrome (MCNS)	D. Chronic Kidney Disease (CKD)
E. Acute Post-Streptococcal Glomerulonephritis (APSGN)	
Clinical Description / Scenario	[Match]
1. An 8-year-old child presenting with cola-colored urine, mild to moderate edema, and hypertension 1-2 weeks after an episode of pharyngitis.	[]
2. A pediatric patient presenting with frothy urine, massive generalized edema (anasarca), hypoalbuminemia (<2.5 g/dL), and severe hyperlipidemia (>250 mg/dL).	[]
3. Persistent abnormalities of kidney structure or function for >3 months, OR a persistent decrease in GFR <60 mL/min/1.73m ² for >3 months.	[]
4. A rapid decline in GFR staged as Stage 2 "Injury" based on pRIFLE criteria, where GFR decreases by 50% or urine output is <0.5 mL/kg/h for 16 hours.	[]
5. A young girl presenting with dysuria, urgency, frequency, and malodorous urine, characterized strictly by "NO FEVER and NO RENAL INJURY".	[]

Group 2: Diagnostics, Pathology & Imaging (Nephrology)

A. Minimal Change Disease (MCNS)	B. DNase B Antibody Titer
C. Voiding Cystourethrogram (VCUG)	D. APSGN Histopathology
E. DMSA Renal Scanning	
Clinical Description / Scenario	[Match]
1. Immunofluorescence pathology showing characteristic "Lumpy bumpy deposits of immunoglobulin and complement on the GBM".	[]
2. Electron microscopy demonstrating "Effacement of epithelial cell foot processes" in a child with suspected Minimal Change Disease (MCNS).	[]
3. The rising antibody titer that serves as the "best tool for cutaneous (skin) streptococcal evidence" evaluation.	[]
4. A nuclear imaging modality designated as "useful if acute pyelonephritis is uncertain or to assess SCARRING".	[]
5. A contrast imaging study indicated in "ALL children <5 years old with a febrile UTI" to screen for Vesicoureteral Reflux (VUR).	[]

Group 3: Treatment, Management & Lab Differentiation (Nephrology)

A. Intrinsic Renal AKI Differential Diagnostic Profile	B. Management of Proteinuria & Hypertension in CKD
C. Pre-Renal AKI Differential Diagnostic Profile	D. Supportive Treatment in Nephrotic Syndrome
E. Specific Measures in APSGN Management	

Clinical Description / Scenario	[Match]
1. Administration of IV 25% Human Albumin (0.5 g/kg) with Furosemide, indicated for massive anasarca causing respiratory distress or hypovolemic shock.	[]
2. A 10-day course of oral penicillin to eradicate active streptococci, noting that it "does not alter the GN natural history".	[]
3. Long-term administration of ACE inhibitors (such as enalapril or lisinopril) specifically indicated for managing proteinuria and hypertension.	[]
4. A diagnostic lab profile showing: Urine Specific Gravity >1020, Osmolality >500, UNa <20 mEq/L, and FENa <1%.	[]
5. A diagnostic lab profile showing: Urine Specific Gravity <1010, Osmolality <350, UNa >40 mEq/L, and FENa >2%.	[]

Group 4: Neonatal Jaundice & Bilirubin Toxicity (Neonatology)

A. Phase 2 of Acute Bilirubin Encephalopathy	B. Chronic Bilirubin Encephalopathy (Kernicterus)
C. Pathological Jaundice	D. Phase 1 of Acute Bilirubin Encephalopathy
E. Physiological Jaundice	

Clinical Description / Scenario	[Match]
1. Jaundice appearing within the first 24 hours of life, with a Total Serum Bilirubin (TSB) rise > 5 mg/dL/day or conjugated bilirubin > 2 mg/dL.	[]
2. Jaundice appearing after 24 hours of life (typically day 2-3), peaking on day 4-5, and resolving spontaneously within 10-14 days.	[]
3. The initial phase of Acute Bilirubin Encephalopathy (ABE) characterized clinically by lethargy, hypotonia, and a poor suck reflex.	[]
4. The intermediate phase of ABE characterized by irritability, hypertonia, a high-pitched cry, retrocollis, and opisthotonos.	[]
5. Chronic permanent clinical sequelae of bilirubin toxicity presenting with athetoid cerebral palsy, sensorineural hearing loss, and dental enamel dysplasia.	[]

Group 5: Neonatal Sepsis & Complications of Prematurity (Neonatology)

A. Late-Onset Sepsis (LOS)	B. Respiratory Distress Syndrome (RDS)
C. Patent Ductus Arteriosus (PDA)	D. Early-Onset Sepsis (EOS)
E. Necrotizing Enterocolitis (NEC)	

Clinical Description / Scenario	[Match]
1. Infection manifesting within < 72 hours of birth, typically associated with maternal risk factors and vertical transmission of GBS or E. coli.	[]
2. Infection manifesting > 72 hours after birth, typically nosocomial (hospital-acquired) or community-acquired, commonly caused by CoNS or S. aureus.	[]
3. A prematurity complication due to surfactant deficiency, presenting with grunting, tachypnea, and a classic "ground-glass" opacity on chest X-ray.	[]
4. A severe gastrointestinal complication in preterm infants presenting with abdominal distension, bloody stools, and "pneumatosis intestinalis" on X-ray.	[]
5. A cardiovascular complication of prematurity where incomplete closure of the shunt causes a continuous machine-like murmur and bounding pulses.	[]

Group 6: Transient Cutaneous Lesions of the Newborn (Neonatology)

A. Erythema Toxicum Neonatorum	B. Neonatal Acne (Cephalic Pustulosis)
C. Milia	D. Seborrheic Dermatitis
E. Mongolian Spots (Congenital Dermal Melanocytosis)	

Clinical Description / Scenario	[Match]
1. Tiny, 1-2 mm pearly white or yellowish keratin-filled cysts commonly clustered over the nose, chin, and forehead, resolving within weeks.	[]
2. A benign rash appearing on day 2-3 of life as erythematous macules and papules containing central pustules filled predominantly with eosinophils.	[]
3. Flat, blue-gray melanocytic macules or patches typically located over the lumbosacral region, common in darker-skinned infants, and fade with time.	[]
4. Inflammatory comedones, papules, and pustules appearing around 2-3 weeks of life primarily on the face, induced by maternal androgens.	[]
5. Blotchy erythematous patches with fine scaling often found on the scalp, eyebrows, and behind ears, commonly called "cradle cap".	[]